

# Quantifying incident impacts and identifying influential features in urban traffic networks

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## INTRODUCTION

### Summary

- Formulated a new nonrecurrent congestion measures incident duration using trajectory-based average road speeds.
- These measures can be applied without accurate time information of incidents and assess both temporal and spatial impacts.
- Applied LIME technique to develop prediction models and extract meaningful features related to the proposed congestion measures.

### Problem Statement

- To get interpretability for traffic congestion impact, it's formulated to extract feature importance  $X \in R^{N \times C}$  by train a function  $f_m(\cdot)$  that maps a feature matrix  $X$  to incident impact level,  $y^m$ .

$$\mathbf{X} \xrightarrow{f_m(\cdot)} \mathbf{y}^{(m)}$$

## METHODOLOGY

### Measuring nonrecurrent traffic congestion

#### Incident Duration

$$t_{v0}^{st} = \min\{t: t \text{ satisfies } (c1), (c2), (c3)\}, t = \dots, t_{v0}^{rpt} - 1,$$

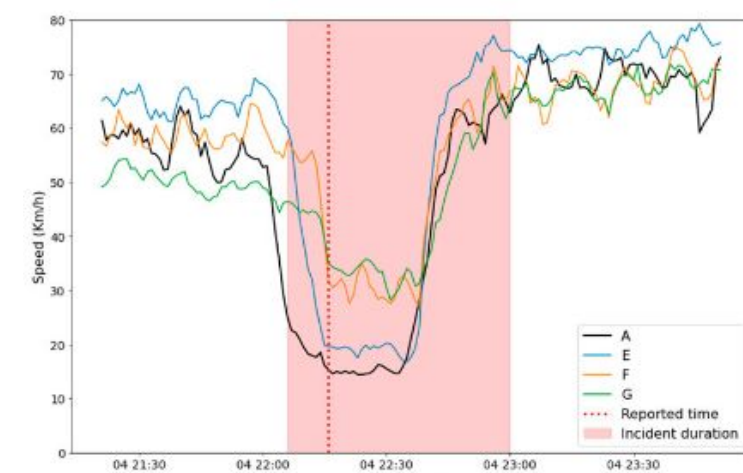
$$(c1): \bar{s}(v_0, t)_5 \leq S^{REG}(v_0, t)$$

$$(c2): \bar{s}(v_0, t)_5 \leq \bar{s}(v_0, t)_{30}$$

$$(c3): s(v_0, t) \leq s(v_0, t-1) - sd(s(v_0, [t-30min:t]))$$

$$t_{v0}^{end} = \min\{t: S^{REG}(v_0, t) < avg(s(v_0, [t-30min:t]))\}, t = t_{v0}^{end} + \delta + 1, \dots$$

$$(Incident \text{ Duration}) y_{v0}^1 := t_{v0}^{end} - t_{v0}^{st}$$

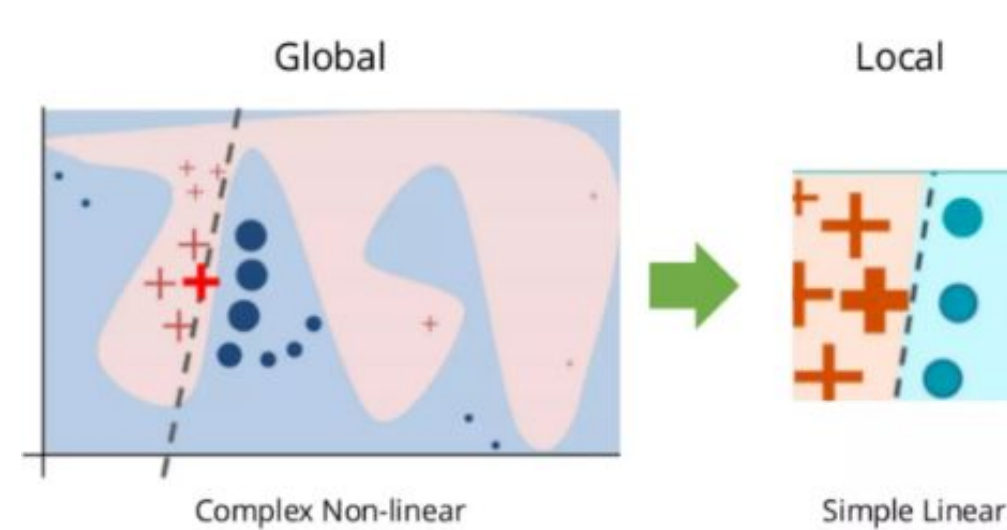


### Model-based feature importance: LIME(Local Interpretable Model-Agnostic Explanations)

- Objective: find the tangent plane to the ML surface, in the point  $(y_i, X_i)$  we want to explain.
- Solution: Sample points on the ML surface, approximate the tangent with a linear model through the points (Ridge Regression), in the neighborhood of the reference individual.

$$\epsilon(X) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \pi_x) + \omega(g)$$

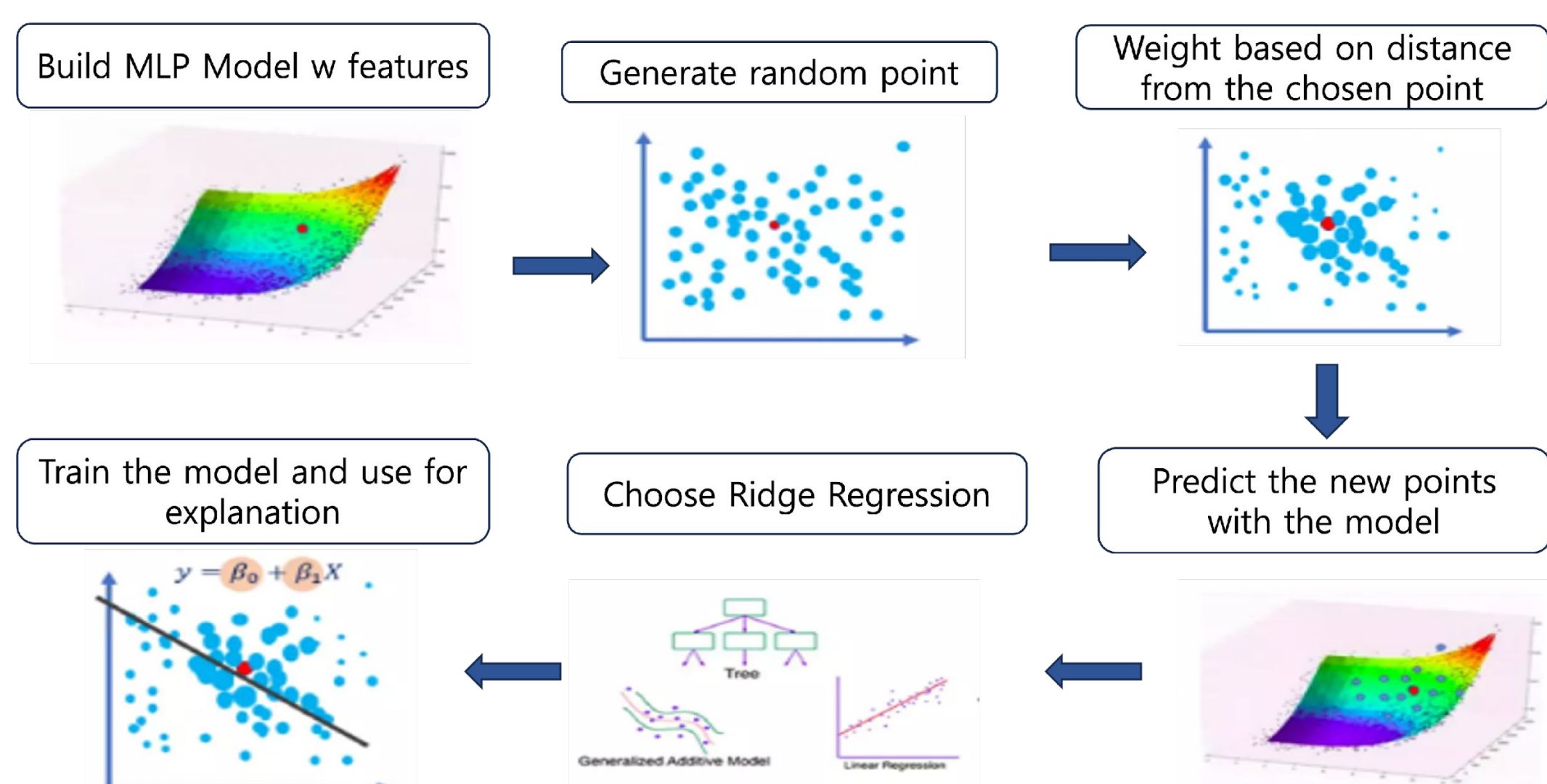
where  $\mathcal{L}(f, g, \pi_x)$  is a measure of how unfaithfully  $g$  approximates  $f$  in the locality defined by  $\pi_x$ , and  $\omega(g)$  is a measure of the complexity of the explanation  $g \in G$ .



### Algorithm Summary

- Choose the ML model and reference point to be explained
- Generate points all over the  $\mathbb{R}^p$  space
- Predict the Y coordinate of the sampled points, using the ML model
- Assign weights based on the closeness to the chosen point
- Train Linear Ridge Regression on the generated weighted dataset: The  $\beta$  coefficients are regarded as LIME explanation.

$$E(Y) = \beta_0 + \sum \beta_j X_j$$



## EXPERIMENT

### Feature Importance Measure

- $h_{i,j}$  be the coefficient corresponding to the j-th feature in  $\epsilon(X_i)$
- $N$  is the number of incidents
- $I_j$  is feature importance
- $RI_j$  is a relative contribution of each feature

$$I_j = \frac{\sum_{i=1}^N |h_{i,j}|}{N}$$
$$RI_j = \frac{I_j}{\sum_{j=1}^C I_j}$$

### Features

- Key independent features extracted from both the traffic dataset and incident dataset for the incident-anchored subgraphs.
- The extracted features can be categorized as follows: Timestamp, Road Characteristics, Network Topology, Speed, Source, and Weather.

| Categories           | Features                  | Value set & Description                                                                                                                                            |
|----------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Timestamp            | time of day               | 1: 0 ≤ incident time < 6<br>2: 6 ≤ incident time < 12<br>3: 12 ≤ incident time < 18<br>4: 18 ≤ incident time < 24<br>0: weekday<br>1: weekend (Saturday or Sunday) |
|                      | day of week               | 1: highway<br>2: urban highway<br>3: general national road<br>4: state-sponsored regional road<br>5: local road<br>6: general road                                 |
| Road Characteristics | road type                 | 30 ≤ s_max(v_0) < 120<br>length of the road segment where the incident occurred<br>number of lanes of an incident road                                             |
|                      | incident road speed limit | $ V_{v0}^{in} $                                                                                                                                                    |
| Network topology     | incident road length      | $ V_{v0}^{out} $                                                                                                                                                   |
|                      | number of lanes           | $Cd_{in}$                                                                                                                                                          |
| Speed                | number of incoming roads  | $Cd_{out}$                                                                                                                                                         |
|                      | number of outgoing roads  | $Cc$                                                                                                                                                               |
| Source               | in-degree centrality      | $Cb$                                                                                                                                                               |
|                      | out-degree centrality     | $Ce$                                                                                                                                                               |
| Weather              | closeness centrality      | $corr_{in}(v_0)$                                                                                                                                                   |
|                      | betweenness centrality    | $corr_{out}(v_0)$                                                                                                                                                  |
| Speed                | eigenvector centrality    | $0 \leq \rho_{eff} \leq 1$                                                                                                                                         |
|                      | incoming correlation      | standard deviation of speed data prior to incident occurrence                                                                                                      |
| Source               | outgoing correlation      | 1: traffic broadcasting<br>2: national transport information center                                                                                                |
|                      | prior speed drop ratio    | 3: police station                                                                                                                                                  |
| Weather              | speed variation           | 4: infrastructure and transport                                                                                                                                    |
|                      | information source        | 0: clear weather (sunny)<br>1: cloudy<br>2: rain<br>3: rain amount<br>4: rain probability<br>5: temperature                                                        |

## RESULT

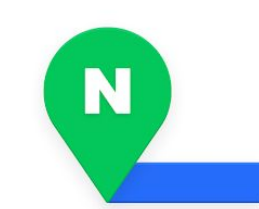
### Feature Importance

- 'Incoming correlation' is the most influential factor for incident duration with a relative importance of 12.46%.
- 'Incident road length' also significantly contributes, holding a 7.56% relative importance.
- These factors are strongly linked to the traffic capacity of the incident road and inflow traffic volume, with inflow impacting duration speed drop more than outgoing traffic.
- The weather code, an environmental incident factor, has a 6.96% relative importance on incident duration.

| Categories           | Features                  | Incident duration |              |
|----------------------|---------------------------|-------------------|--------------|
|                      |                           | $I_j$             | $RI_j(\%)$   |
| Timestamp            | time of day               | 5.716             | 7.21         |
|                      | day of week               | 3.249             | 4.10         |
| Road characteristics | road type                 | 1.649             | 2.08         |
|                      | incident road speed limit | 2.983             | 3.76         |
| Network topology     | incident road length      | 5.998             | 7.56         |
|                      | number of lanes           | 3.090             | 3.90         |
| Speed                | number of incoming roads  | 2.560             | 3.23         |
|                      | number of outgoing roads  | 2.577             | 3.25         |
| Source               | in-degree centrality      | 2.788             | 3.52         |
|                      | out-degree centrality     | 2.429             | 3.06         |
| Weather              | closeness centrality      | 2.471             | 3.12         |
|                      | betweenness centrality    | 2.965             | 3.74         |
| Speed                | eigenvector centrality    | 2.184             | 2.75         |
|                      | incoming correlation      | <b>9.880</b>      | <b>12.46</b> |
| Source               | outgoing correlation      | 2.601             | 3.28         |
|                      | prior speed drop ratio    | 2.217             | 2.80         |
| Weather              | volatility                | 3.366             | 4.24         |
|                      | information source        | 3.969             | 5.00         |
| Weather              | weather code              | 5.517             | 6.96         |
|                      | rain amount               | 4.257             | 5.37         |
| Weather              | rain probability          | 2.820             | 3.56         |
|                      | temperature               | 4.015             | 5.06         |

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## REFERENCES

- Afrin, Tanzina, and Nita Yodo. 2020. "A Survey of Road Traffic Congestion Measures Towards a Sustainable and Resilient Transportation System." *Sustainability* 12 (11): 4660.
- Al-Deek, H., A. Garib, and A. E. Radwan. 1995. "New Method for Estimating Freeway Incident Congestion." *Transportation Research Record* (1494): 30-39.
- Ausloos, M., and K. Ivanova. 2002. "Mechanistic Approach to Generalized Technical Analysis of Share Prices and Stock Market Indices." *The European Physical Journal B-Condensed Matter and Complex Systems* 27 (2): 177-187.